

Integration WS 7

Evaluate Q1-20

1. $\int \operatorname{cosec} x \, dx \quad \left(t = \tan \frac{x}{2}\right)$

2. $\int \sec x \, dx \quad \left(t = \tan \frac{x}{2}\right)$

3. $\int_0^1 \frac{x}{x^4+1} \, dx \quad (u = x^2)$

4. $\int_4^9 \frac{1}{(x-1)\sqrt{x}} \, dx \quad (u = \sqrt{x})$

5. $\int_2^6 x\sqrt{6-x} \, dx \quad (u^2 = 6-x)$

6. $\int_{\frac{1}{\sqrt{3}}}^1 \frac{1}{x^2\sqrt{1+x^2}} \, dx \quad (x = \tan \theta)$

7. $\int_0^1 \frac{\sqrt{x}}{1+x} \, dx \quad (x = \tan^2 \theta)$

8. $\int_0^2 \sqrt{\frac{x}{4-x}} \, dx \quad (x = 4 \sin^2 \theta)$

9. $\int_0^{\frac{\pi}{3}} \frac{1}{1-\sin x} \, dx \quad \left(t = \tan \frac{x}{2}\right)$

10. $\int_0^{\frac{\pi}{2}} \frac{1}{3+5 \cos x} \, dx \quad \left(t = \tan \frac{x}{2}\right)$

11. $\int \frac{dx}{2(1+x)\sqrt{x}}$

12. $\int \frac{x}{x^4+1} \, dx$ (Using $u = x^2$)

13. $\int \frac{dx}{1+e^x}$

14. $\int \frac{1+e^x}{1-e^x} \, dx$ (Hint $t = e^x$)

15. $\int_2^3 x^2 \sin x \, dx$

16. $\int x^2 \ln x \, dx$

17. $\int x\sqrt{1-x} \, dx$

18. $\int \frac{dx}{\sqrt{1+x^2}}$

19. $\int \frac{dx}{x\sqrt{x^2-1}}$

20. $\int \frac{e^x}{1+e^{2x}} \, dx$

21. If $I_n = \int_0^{\frac{\pi}{2}} \frac{\cos(2n+1)\theta}{\cos \theta} \, d\theta$, show that $I_n +$

$I_{n-1} = 0$ for $n \geq 1$. Hence find the value of I_n for $n \geq 0$.

22. If $I_n = \int_0^a (a^2 - x^2)^n \, dx$ for $n \geq 0$, show that

$$I_n = \frac{2a^2n}{2n+1} I_{n-1} \text{ for } n \geq 1.$$

23. If $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \cos^2 x \, dx$ for $n \geq 0$, show

$$\text{that } I_n = \frac{n-1}{n+2} I_{n-2} \text{ for } n \geq 2. \text{ Hence show}$$

$$\text{that } \int_0^{\frac{\pi}{2}} \sin^4 x \cos^2 x \, dx = \frac{\pi}{32}$$

24. If $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx$ for $n \geq 0$, show that

$$I_n = n \left(\frac{\pi}{2}\right)^{n-1} - n(n-1)I_{n-2} \text{ for } n \geq 2.$$

$$\text{Hence evaluate } \int_0^{\frac{\pi}{2}} x^4 \sin x \, dx.$$

25. If $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$ for $n \geq 0$, show that

$$I_n = \frac{n-1}{n} I_{n-2} \text{ for } n \geq 2. \text{ Hence find } I_{10} \text{ and}$$

use a suitable change of variable to find

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos^{10} 2\theta \, d\theta$$

26. Show that $\int_0^a f(x) \, dx = \int_0^a \{f(x) +$

$$f(a-x)\} \, dx. \text{ Hence show that}$$

$$\int_0^{\pi} x \sin^6 x \, dx = \pi \int_0^{\frac{\pi}{2}} \sin^6 x \, dx = \frac{5\pi^2}{32}$$

27. a. i. Find the values of A and B such that

$$\frac{3x^2-10}{x^2-4x+4} = 3 + \frac{A}{x-2} + \frac{B}{(x-2)^2}$$

ii. Hence find $\int \frac{3x^2-10}{x^2-4x+4} \, dx$

b. i. Find the integer A, B, C and D such that

$$\frac{3x+7}{(x-1)^2(x-2)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x-2} + \frac{D}{(x-2)^2}$$

ii. Hence find $\int \frac{3x+7}{(x-1)^2(x-2)^2} \, dx$

c. Show that: $\int_4^6 \frac{x^2-8}{x^3+4x} \, dx = \frac{3}{2} \ln 2 - 2 \ln \frac{3}{2}$

28. a. $\int_0^2 \frac{1+4x}{(4-x)(x^2+1)} dx = \frac{1}{2} \ln 20$ Let $\frac{x^2-1}{x^4+x^2} = \frac{A}{x} +$

$\frac{B}{x^2} + \frac{Cx+D}{x^2+1}$. Find A, B, C and D .

b. Hence show that $\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{x^2-1}{x^4+x^2} dx = \frac{1}{3}(\pi - 2\sqrt{3})$

29. Use appropriate methods to find:

a. $\int \frac{x^2+1}{x^2-1} dx$

b. $\int \frac{x^2+1}{x^2-x} dx$

c. $\int \frac{x^3+1}{x^3+x} dx$

d. $\int \frac{x^2}{x^2-5x+6} dx$

e. $\int \frac{x^3+5}{x^2+x} dx$

f. $\int \frac{x^4}{x^2-3x+2} dx$

30. Use a similar approach to Question 29 for repeated factors to show that:

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{(5x-x^2)}{(x+1)^2(x-1)} dx = 4 - 3 \ln 3$$

31. Let $I_n = \int \frac{x^n}{\sqrt{x^2-a^2}} dx$, show that $nI_n -$

$$(n-1)a^2 I_{n-2} = x^{n-1} \sqrt{x^2-a^2}. \text{ Hence}$$

$$\text{evaluate } \int_{2a}^{3a} \frac{x^4}{\sqrt{x^2-a^2}} dx$$

32. $I_n = \int_0^1 x^n (1-x)^n dx, n = 0, 1, 2$

a. Using the substitution $u = \frac{1}{2} - x$, show

that $I_n = \int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\frac{1}{4} - u^2\right)^n du$ and hence

$$I_n = \frac{n}{2(2n+1)} I_{n-1} \text{ for } n = 1, 2, 3$$

b. Show that $\int_0^1 x^5 (1-x)^5 dx = \frac{(5!)^2}{11!}$

c. Use the substitution $x = \sin^2 \left(\frac{1}{2}t\right)$ to show that $\int_0^\pi \sin^{2n+1} t dt = 2^{2n+1} I_n$.

Hence deduce that $\int_0^\pi \sin^{2n+1} t dt = \frac{2^{2n+1}(n!)^2}{(2n+1)!}$ For $n = 0, 1, 2 \dots$

33. a. Express $x^2 + 2x + 5$ in the form

$$(x+a)^2 + b^2. \text{ Hence evaluate}$$

$$\int_{-1}^1 \frac{1}{x^2+2x+5} dx$$

b. Find $\int \frac{\ln x}{\sqrt{x}} dx$

c. i. Show that $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{1}{x(\pi-2x)} dx = \frac{2}{\pi} \ln 2$

ii. Hence use the substitution $u = \frac{\pi}{2} - x$ to

evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cos^2 x}{x(\pi-2x)} dx$

Answers

1. $\ln \left| \tan \frac{x}{2} \right|$

2. $\ln \tan \left(\frac{x}{2} + \frac{\pi}{4} \right)$

3. $\frac{\pi}{8}$

4. $\ln \frac{3}{2}$

5. $\frac{96}{5}$

6. $2 - \sqrt{2}$

7. $2 - \frac{\pi}{2}$

8. $\pi - 2$

9. $1 + \sqrt{3}$

10. $\frac{1}{4} \ln 3$

11. $\tan^{-1} \sqrt{x}$

12. $\frac{1}{2} \tan^{-1} x^2$

13. $x - \ln(1 + e^x)$

14. $x - 2 \ln(1 - e^x)$

15. 3.3

16. $\frac{x^2}{3} \ln x - \frac{x^3}{9}$

17. $\frac{2}{5}(1+x)^{\frac{5}{2}} - \frac{2}{3}(1+x)^{\frac{3}{2}}$

18. $-\frac{\sqrt{1+x^2}}{x}$

19. $\sec^{-1} x$

20. $\tan^{-1} e^x$

21. $(-1)^n \frac{\pi}{2}$

24. $\frac{1}{2}(\pi^3 - 24\pi + 48)$

25. $\frac{63\pi}{512}, \frac{63\pi}{128}$

27. a. i. $A = 12, B = 2$, ii. $3x +$

$12 \ln(x-2) - \frac{2}{x-2} + C$; b. i.

$A = 23, B = 10, C = -23, D = 13,$

ii. $23 \ln \left(\frac{x-1}{x-2} \right) - \frac{10}{x-1} - \frac{13}{x-2} + C$

28. a. $A = 0, B = -1, C = 0, D = 2$

29. a. $x + \ln(x-1) - \ln(x+1) + C$; b.

$x + 2 \ln(x-1) - \ln x + C$; c.

$x - \tan^{-1} x + \ln x - \frac{1}{2} \ln(x^2+1) +$

C ; d. $x + 9 \ln(x-3) -$

$4 \ln(x-2) + C$; e. $\frac{1}{2}x^2 - x +$

$5 \ln x - 4 \ln(x+1) + C$; f.

$\frac{1}{3}x^3 + \frac{3}{2}x^2 + 7x + 16 \ln(x-2) -$

$\ln(x-1) + C$

31. $\frac{27a^4\sqrt{2}}{2} - 2a^4\sqrt{3} + \frac{9a^4\sqrt{2}}{4} - \frac{3a^4\sqrt{3}}{4} +$

$\frac{3a^4}{8} \ln \left(\frac{3+2\sqrt{2}}{2+\sqrt{3}} \right)$

33. a. $a = 1, b = 2, \frac{\pi}{8}$; b. $2\sqrt{x} \ln x -$

$4\sqrt{x}$; c. $\frac{\ln 2}{\pi}$